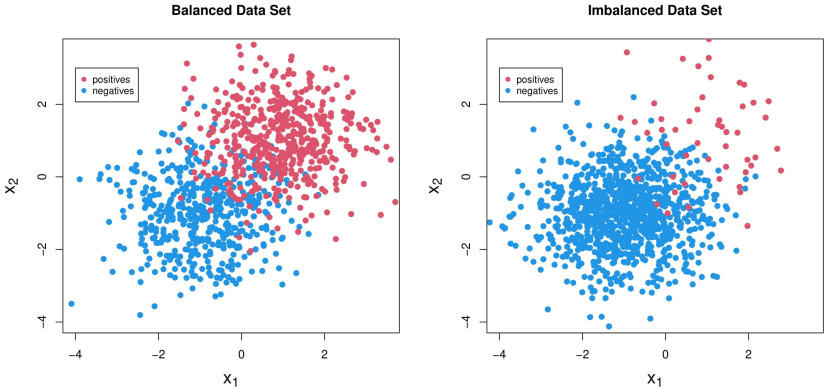
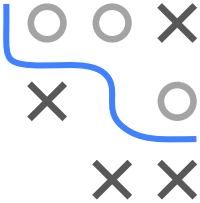
# IMBALANCED DATA SETS

- Class imbalance: Ratio of classes is significantly different.
- Consequence: Undesirable predictive behavior for smaller class.
- Example: Sampling from two Gaussian distributions



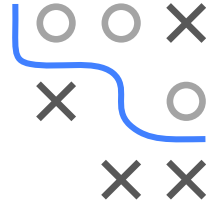
# IMBALANCED DATA SETS: EXAMPLES



| Domain | Task | Majority Class | Minor Class |
|---|---|---|---|
| Medicine | Predict tumor pathology | Benign | Malignant |
| Information retrieval | Find relevant items | Irrelevant items | Relevant items |
| Tracking criminals | Detect fraud emails | Non-fraud emails | Fraud emails |
| Weather prediction | Predict extreme weather | Normal weather | Tornado, hurricane |

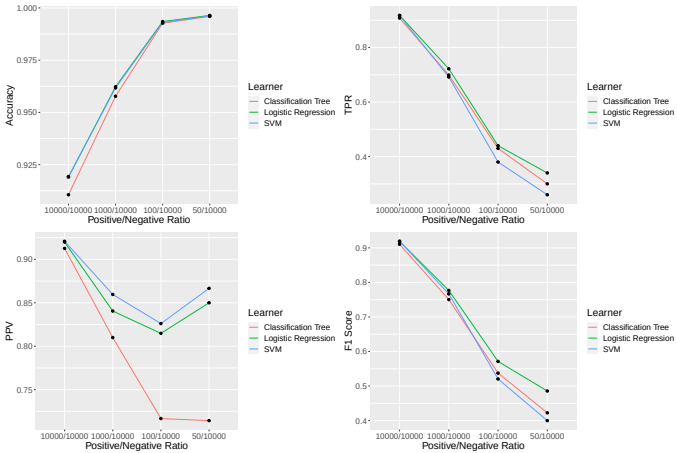
- Often, the minority class is the more important class.
- Imbalanced data can be a source of bias related to concept of fairness.

# ISSUES WITH EVALUATING CLASSIFIERS



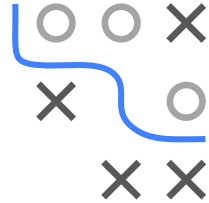
- Ideal case: correctly classify as many instances as possible
  $\Rightarrow$ High accuracy, preferably 100%.
- In practice, we often obtain on imbalanced data sets: **good** performance on the **majority** class(es), a **poor** performance on the **minority** class(es).
- Reason: the classifier is biased towards the **majority** class(es), as predicting the majority class pays off in terms of accuracy.
- Focusing only on accuracy can lead to bad performance on minority class.
- Example:
  - Assume that only 0.5% of the patients have a disease,
  - Always predicting “no disease” leads to accuracy of 99.5%

# ISSUES WITH EVALUATING CLASSIFIERS



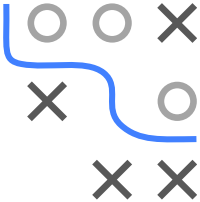

In each scenario, we have 10.000 obs in the negative class. Number of obs in positive class varies between 10.000, 1.000, 100, and 50. Train classifiers with 10-fold stratified cv. Evaluate via aggregated predictions on test set.

# POSSIBLE SOLUTIONS



- Ideal performance metric: the learning is *properly* biased towards the minority class(es).
- Imbalance-aware performance metrics:
  - G-score
  - Balanced accuracy
  - Matthews Correlation Coefficient
  - Weighted macro $F_1$ score

# POSSIBLE SOLUTIONS



| Approach | Main idea | Remark |
|---|---|---|
| Algorithm-level | Bias classifiers towards minority | Special knowledge about classifiers is needed |
| Data-level | Rebalance classes by resampling | No modification of classifiers is needed |
| Cost-sensitive Learning | Introduce different costs for misclassification when learning | Between algorithm- and data-level approaches |
| Ensemble-based | Ensemble learning plus one of three techniques above | - |